

Setting Up the Program's Constants

At the beginning of the source code, the program uses a few constants to make the code more readable. The Python code is as follows:

Python

```
import random, time

DIFFICULTY = 40 # How many random slides a puzzle starts with.
SIZE = 4 # The board is SIZE x SIZE spaces.
random.seed(1) # Select which puzzle to solve.

BLANK = 0
UP = 'up'
DOWN = 'down'
LEFT = 'left'
RIGHT = 'right'
```

The JavaScript code is as follows:

JavaScript

```
<script type="text/javascript">
const DIFFICULTY = 40; // How many random slides a puzzle starts with.
const SIZE = 4; // The board is SIZE x SIZE spaces.

const BLANK = 0;
const UP = "up";
const DOWN = "down";
const LEFT = "left";
const RIGHT = "right";
```

To have reproducible random numbers, the Python program sets the random number seed to 1. The same seed value will always reproduce the same random puzzle, which is useful for debugging. You can change the seed value to any other integer to create different puzzles. JavaScript has no way to set its random seed value, and *slidingtilesolver.html* doesn't have an equivalent feature.